

US 29794 v2a

Assessment Sample Answers

GENERIC COMPUTING

Unit Standard 29794

Version 2 | Level 3 | Credits 5

Describe risks and implement security solutions when using digital tools

Important Note for Assessors:

1. These sample answers and/or assessor's comments are supplied for the use of assessors and moderators only.
2. A copy of this **Assessment Sample Answers** document and any other assessment evidence must be submitted with the **Assessment Booklet** when moderation is called for.
3. Sample answers are indicative only and learner's answers may differ from the sample answers, provided they meet the requirements of the tasks. The learner's responses should be reasonably comprehensive as judged by the Assessor for level 3.

Level 3 Descriptor

KNOWLEDGE

Some operational and theoretical knowledge in a field of work or study.

SKILLS

Select and apply from a range of known solutions to familiar problems.

Apply a range of standard processes related to the field of work or study.

APPLICATION

[OF KNOWLEDGE AND SKILLS]

Limited supervision.

Requiring major responsibility for own learning and performance.

Adapting own behaviour when interacting with others.

Contributing to group performance.

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Note to Assessor:

Task 1 and Task 2 Part 1 may be done as a presentation, project or report if preferred. If that is the case, please use the following model answers as a guide to ensure the requirements are met.

Assessment Task 1: Legal and regulatory risks and compliance requirements

Outcome 1, PC 1.1 to 1.3

Privacy and security

What is meant by security in terms of data storage and transmission?

Ensuring unauthorised individuals or organisations do not access the data.

What legislation and/or regulations apply to the privacy of an individuals data that may be stored or transmitted by an organisation.

The Privacy Act 2020 sets rules for when and why personal information can be collected and shared, and how it should be stored.

What are the risks to digital security and privacy of information that an organisation may face? (Consider device use and storage and transmission of information).

If there is a security breach the organisation may lose access to its data, for example it may be ransomed. Their competitors may also access commercially sensitive information.

For organisations that store personal information, there is the risk of a privacy (data) breach, which is when someone's personal information is lost, or is accessed by an unauthorised person. This can have serious implications for individuals, such as identity theft (eg where someone uses someone else's personal information to pretend to be that person).

A data breach could also have a negative impact on the organisation's reputation. People may lose trust in the organisation, and not want to use their services or buy anything from them in future.

What are the implications for the organisation with regards to its data, systems and processes?

The Privacy Act requires organisations to take all reasonable steps to keep personal information secure and to prevent unauthorised access.

Individuals may lay a complaint with the Privacy Commissioner against organisations that do not take all reasonable steps to secure and ensure the accuracy of personal information they hold. The Commissioner may take court proceedings against the organisation and this may result in the organisation needing to compensate the individual for damages resulting from a data breach (eg a financial settlement).

Organisations need to have clear policies and procedures concerning the secure storage and transmission of data.

How does your organisation comply with data security and privacy requirements? Where available, consider policies such as access control and acceptable use.

Examples may include:

- *the use of passwords*
- *data encryption*
- *logging out when not at their computer*
- *training around acceptable use and access and sharing of data*
- *data validation policies to ensure data is accurate*
- *anti-malware measures*
- *handling of devices such as devices and media taken offsite must not be left unattended in public places and not left in sight in a car.*

Many organisations include a Privacy Statement on their website and/or communications they have with their clients. This indicates how the organisation collects, stores, and uses any personal information they collect.

Copyright

Describe what is meant by copyright. Include any relevant legislation or regulations in your answer.

In New Zealand, any original work is automatically subject to copyright (ie the 'right to copy'). A person (or organisation) who holds the copyright for an item has exclusive rights to copy, publish, distribute, adapt, etc their work. They also have the exclusive right to give up the copyright to their work (eg by releasing it under a Creative Commons licence or into the public domain).

The ***Copyright Act 1994*** *sets out the copyright restrictions that apply in New Zealand. The* ***Copyright (New Technologies) Amendment Act 2008*** *clarifies the definition of 'copying' and 'copy' so that it clearly includes making digital copies of works.*

What are some possible issues relating to copyright that an organisation may face?

An organisation may wish to use a copyrighted work in a publication, for example, an advertisement. If they do not have the correct licence to use the work they could be liable for legal action eg a fine or even jail. A director, manager, and/or owner of a business can be held responsible if they knowingly allow their employees to break copyright or do not take steps to prevent their employees breaking copyright.

What are the implications for the organisation with regards to its data, systems and processes?

Organisations must:

- *provide training and education related to copyright to all members of the organisation, to help ensure they do not intentionally or unintentionally break copyright requirements*
- *have in place policies and procedures related to copyright, including the specific copyright requirements that apply to that particular organisation*

How does your organisation comply with issues related to copyright? Where available, consider policies such as acceptable use.

In our organisation we have a subscription to a stock photo service for our website and other advertising material. This gives us a licence to use those photos for commercial purposes.

Spamming

What is spamming?

Spam is any unsolicited electronic message (ie a message you did not ask for). This includes e-mail, instant messaging, SMS, and mobile phone messaging.

What is the applicable legislation?

*The **Unsolicited Electronic Messages Act 2007** prohibits spam being sent to, from, or within New Zealand.*

How does your organisation comply with this legislation?

The following are examples of steps organisations may take to ensure they comply with the Act.

1. ***Get consent from the recipient.*** *The person needs to give their permission to be sent commercial messages.*
2. ***Clearly identify themselves.*** *The message needs to clearly indicate who sent it and what their contact details are.*
3. ***Include an unsubscribe facility.*** *The recipient needs to be given the option to unsubscribe at any time. In other words, they need to be able to say if they don't want commercial messages sent to them in the future. If someone asks to 'unsubscribe', this request must be actioned within 5 working days.*

Software Licencing

What legislation does an organisation have to follow with regards to the software that they use?

Software, like any other item created or developed, is subject to copyright. You may be breaking copyright law if you copy or use software without a licence.

It is important to keep in mind that in the case of software, a user does not buy and own the source code itself. They only buy a **licence to use** the software, and are limited by the requirements of the licence in terms of what they can do with the software.

How does your organisation ensure they are using software within the terms of the licence?

Examples include:

- Prohibiting users from installing software unless they have the correct permission eg an admin login
- Including information concerning software and its use in their IT policy

Assessment Task 2: Security measures

Outcome 2, PC 2.1 and PC 2.2

Data interception and unauthorised access

Description of risk:

Data being accessed by someone who does not have permission to do so.

Procedures to address risk and meet good practice:

The end user needs to:

- *use a user ID, passwords and/or passcodes to secure their devices*
- *password protect digital files containing sensitive or personal information*
- *change passwords regularly and never share them with anyone else (either intentionally or unintentionally)*
- *save files containing sensitive or personal information as read-only – to avoid the data being changed or corrupted*
- *log off from their computer (or lock it), if it is going to be unattended*
- *make sure that mobile devices are securely stored when outside of the office (eg not leaving them in plain sight in the car, etc)*
- *shut down their computer when they leave the office*
- *make sure that they switch off their monitor if someone comes close enough to see what is on the screen*
- *make sure that malware and operating system software is up-to-date.*

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Malware

Description of risk:

Malware is malicious software such as a computer virus, Trojan, or worm. These can be used to gain unauthorised access, and/or to corrupt or steal data.

Procedures to address risk and meet good practice:

The end user needs to:

- make sure that malware and operating system software is up-to-date
- run regular malware checks, including scans of devices, emails, and files
- never switch off/disable the antimalware tool
- be extremely cautious when receiving emails from an unknown sender (eg not clicking on any links in the email, or opening attachments)
- only download and install software from the Internet that is from a trusted source
- not use public computers or an unsecured public internet connection when entering or sending personal or confidential information
- be aware of the symptoms of a malware infection (eg emails sent that you did not write; device crashing and rebooting frequently; missing or corrupted files; device is slower than usual)
- set junk email settings to 'high'.

If a malware infection is suspected, the end user should immediately do the following:

Disconnect their device from any network it is on.

- Run their anti-malware software to do a full malware scan.
- Immediately inform their supervisor that their device may be infected.

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Natural disaster

Description of risk:

Natural disasters are events such as fire, flood, earthquake, or tornado. They can cause damage to your digital devices, and this may result in the data on the devices being lost or corrupted.

Procedures to address risk and meet good practice:

Our organisation has a detailed management plan in place for backing-up and restoring data. This includes the different types of backup the end user needs to do, and when these need to be done. For example, the end user has to set all files (eg Word, Excel, etc) to auto-backup (locally) every 5 minutes. Backups also need to be made to the local server hourly (this has been set up by IT). These backups are stored offsite weekly. The advantage of having the data backed up offsite, is that it's safe from theft, fire, and other local disasters. Certain data also needs to be backed up to cloud servers on an ongoing basis (auto-sync). If the end user becomes aware of any problems with files auto-syncing, they need to immediately inform their supervisor.

All users in the organisation have to be aware of these backup procedures, understand how to implement them, and ensure they implement them on a regular basis.

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Data corruption

Description of risk:

Data corruption refers to unintended changes made to original data when it is stored, processed, transmitted, etc.

Procedures to address risk and meet good practice:

Data corruption can have several different causes and our organisation has procedures in place to prevent data corruption, depending on the possible cause.

- ***Malicious intent*** *– this is when someone sets out to change or damage the data on purpose (eg using malware). The procedures related to the prevention of a malware infection would therefore apply here.*
- ***Unexpected program exits****, for example, as a result of a power failure, power surge, or computer 'crashing'. Our organisation requires that users must only plug all digital tools into a surge protector – to protect against a power surge as a result of, for example, a lightning strike. All PCs are also plugged into a UPS. If a power failure occurs, the UPS gives the PCs a power supply for a few minutes, giving users a chance to save their files and shut down their computers safely. We also need to regularly back up data to minimise data loss as a result of an unexpected program exit (see previous section).*
- ***Hardware defects*** *(eg a faulty hard drive). The procedures related to addressing* ***hardware failure*** *(see next section) apply here.*
- ***Human error*** *(eg people incorrectly entering, changing, or deleting data). Our organisation give regular training to all end users to help ensure accurate data entry. We also need to save all completed files as read-only, to help ensure that there are no unintentional changes made to files.*

  As part of regular backups, we also save different versions of the same file (versioning). This allows the user to return to an earlier version of the file if the data becomes corrupted. (This is set up by the IT department.)

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Hardware failure

Description of risk:

Hardware failure is when a key hardware component (eg hard drive, motherboard, memory) fails, which means you may not be able to use your device or access the data on it.

Procedures to address risk and meet good practice:

Our organisation has policies and procedures in place to ensure end users do the following:

- Ensure their devices are kept in good condition and regularly checked for potential problems (eg dust built up in the computer box).
- Don't spill liquids on their devices. For example, we are not allowed to consume any food or drink while working on any digital device (including peripherals such as printers).
- Have adequate physical protection for mobile devices (eg smartphone and tablet cases to protect against damage from the device being dropped; water-resistant cases if devices need to be used outdoors; etc).
- Immediately report any strange sounds or behaviour by the device.
- Ask for IT help if we need to move a desktop computer.
- Regularly back up files to an external storage location (eg cloud drive) – in case there is a serious hardware failure and the computer is permanently damaged.

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Network access

Description of risk:

Having computers on a network can open up opportunities for the spread of viruses, unauthorised access, and harm to the network and/or individual computers. Once one computer on the network is infected with a virus, it can spread quickly to other computers on the network and cause huge damage.

Procedures to address risk and meet good practice:

Many of the procedures described in the section on malware also apply to addressing risks related to network access (eg use of anti-malware software).

Our organisation also has a firewall in place to protect computer systems from threats from the Internet.

Furthermore, end users must ensure that all devices (including mobile devices that access the network wirelessly), are running up-to-date versions of their operating systems, web browsers, and security software.

Our organisation provides regular training and information to help ensure end users are aware of the dangers that accessing unprotected browsing of the Internet can have for their own devices, as well as all the other devices on the network. This includes the fact that malware that manages to infect a single device on a network can quickly spread to other devices. This means that not only the data on the first infected device, but on all networked devices, can be accessed and/or corrupted.

Part 2: Use security solutions [PC2.2]

Assessor Checklist

Tick each item the learner has demonstrated or for which they have provided evidence of completion.

NOTE: Evidence may be gather over time from a range of different situations including, for example, the use of digital tools to complete another Unit Standard; the use of digital tools to complete work for an organisation, etc.

Item	✓
1. The learner **maintained security measures** in order to meet the organisation's end user security requirements when using digital tools. Tick all of the following that have been demonstrated.	
• logging off	✓
• shutting down	✓
• physical security	✓
• locking device	✓
• password protection or other form of access control	✓
• saving files as read-only	✓
• other (please specify) _____	
2. The learner ensured the following were updated, as necessary, in keeping with organisational requirements. Tick all that were demonstrated.	
• operating system	✓
• anti-malware	✓
• other (please specify) _____	
3. The learner applied the following anti-malware solutions, in keeping with organisational requirements. Tick all that were demonstrated.	
• anti-malware software	✓
• firewalls	✓
• browser settings	✓
• personal network protection	
• other (please specify) *Refer to assessor comments*	✓

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Item	✓
4. The learner applied the following data protection solutions, in keeping with organisational requirements. Tick all that were demonstrated.	
• making regular and automatic backups of data	✓
• saving work regularly	✓
• using a UPS or surge protector	✓
• other (please specify) _____	
5. All of the above security solutions (items 1 to 4) were demonstrated in keeping with **good practice** (ie learner selected and used the appropriate feature/function to ensure the safe and correct use of the chosen digital tools)	✓
6. All of the above were either observed by the assessor (or verifier), or suitable evidence was supplied (eg screenshots, copies of files with access restrictions, etc).	✓

Assessor Comments (eg give examples of how good practice was applied by learner, etc).

The learner correctly applied the security solutions required by their organisation. The learner consistently selected and used appropriate software and/or hardware features and functions to ensure the safe and correct use of their:

- *desktop PC*
- *mobile phone*
- *tablet*

The learner set up auto-logout for their PC – to log them out automatically when inactive for more than 1 minute. When the learner left their PC for more than a few minutes, they locked it.

The learner's PC was password protected, and the learner used a passcode on their mobile phone, and fingerprint recognition on their tablet. The learner also password-protected several sensitive files, and saved files they did not want anyone to alter as 'read-only'.

Before leaving for the day, the learner shut down their desktop PC.

The organisation automatically updates OS and anti-malware software on a regular basis, and ensures the firewall is in place. The learner made sure that for their personal devices the OS and anti-malware updates were also set to automatically install. In addition to this, the learner also set their Junk mail settings in Outlook to 'high' – as required by the organisation.

The learner only used Google Chrome as their internet browser (as required by the organisation). They also set Google Chrome to auto-download updates, block unwanted pop-ups and plugins, and not to save passwords.

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As per org. requirements, the learner did the following in relation to their use of email:

- Disabled autocomplete addresses for their email.
- Took great care when forwarding email not to include entire email chain (only relevant portion).
- Included the organisations standard disclaimer and confidentiality notice in all emails, as well as the required email signature.

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Assessment Task 3: Health and Safety

Outcome 1, PC 1.3; Outcome 3, PC 2.3

Part One

What legislation does an organisation have to comply with when considering health and safety using digital tools?

The Health and Safety at Work Act 2015 sets out the requirements, duties, and rights related to workplace health and safety. The aim of the Act is to help ensure the health and safety of workers and others (eg customers, clients, visitors, etc) in a workplace.

Briefly describe how your organisation complies with this Act in the context of digital tools?

Examples include policies relating to:

- *Provide and maintain a working environment that is without risks to health and safety. In the context of digital work, this includes risks related to the:*
  - *physical environment (eg equipment and how it is positioned, lighting, ventilation, dust, heat, noise, and risk of physical injury or discomfort)*
  - *psychological environment (eg work-related stress and tiredness).*
- *Provide workers with information, training and/or supervision to ensure they are able to protect themselves (and others) from health and safety risks related to working for the organisation.*
- *Monitor the health of workers and the working environment to prevent future injury or illness. In the context of digital work, this could include:*
  - *ergonomic risks such as strain or fatigue from repetitive motions, poor lighting, etc*
  - *physical risks such as noise, heat, exposure to chemicals (eg from a printer).*

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Health and Safety Hazard 1: Chair and desk incorrect height

Sitting in an uncomfortable position for too long can cause muscle pain and discomfort. For example, a desk that is too high for the person using it, can cause them to sit in an awkward position while working. This can lead to discomfort and strain. A chair that is too low may cause shoulder stiffness, and one that is too high may mean strain on legs if there is no support for the feet.

Simple solutions include an adjustable chair. Both the height and the back of the chair should be able to be adjusted to suit the user. A footrest can also be utilised to support the user's feet and legs.

An adjustable desk may also help. Some people use standing desks as a break from sitting for too long.

Health and Safety Hazard 2 Poor lighting

Poor lighting (ie not light enough, or too bright) can lead to sore and strained eyes, headaches, and even migraines.

Lighting which causes glare or shadows can also cause physical problems for users. For example, if there is light shining directly on the computer screen, this can make it difficult for the user to see what is on the screen. As a result, the user may strain their eyes, or lean forward to try to see what is on the screen. This can cause the following common symptoms of visual discomfort:

- Sore, red, dry, or watery eyes
- Eyes feeling 'gritty' or 'heavy'
- Blurred vision
- Headaches (or even migraines)

Do not have a light source directly in front of you such as a window or you may have too much glare. Also do not have a light source behind you to reflect on the screen. Overhead lights should have diffusers to reduce brightness.

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Health and Safety Hazard 3: *Air quality*

Poor air quality can cause issues with allergies, irritability and stress. Poor ventilation or a room that is too warm can cause headaches and fatigue. Poor ventilation can cause increased illness. If the humidity is too high it can cause stuffiness and fatigue, if the air is too dry it can cause a feeling of dryness and irritation in the nasal passages. It be difficult to be productive under these conditions.

Air temperature should be between 20-26 degrees, depending on the time of year and how heavy people/s clothing is. Humidity depends on temperature - the colder it is, the higher the humidity should be, but never above 80% or below 40%. Rooms should be ventilated and ensure that no-one is sitting in a draught. Air conditioning filters should be kept clean.

Health and Safety Hazard 4: *High noise levels*

Noise levels that are too high can make it difficult to communicate and/or concentrate. A noisy environment can cause mental strain, stress, and fatigue.

Acoustic ceiling tiles, and acoustic partitions around desks can help reduce ambient noise. Having noisy equipment such as printers in their own room can also make a big difference to the amount of noise. It also has the added bonus of forcing people to take small breaks as they have to get up to go to the printer.

Health and Safety Hazard 5: *Poorly laid out work station*

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The way that digital device components – eg the monitor, keyboard, mouse, etc are arranged can have a negative impact on the user's physical health – particularly in terms of causing stress, strain, and discomfort. Below are some examples.

If the device screen is too high or low, the user has to look up or down. This can cause strain and discomfort in the user's neck and shoulders.

If the keyboard is not in line with the user's body, the user may need to twist in an awkward way to type on the keyboard. This can cause strain on the wrists, fingers, neck, shoulders, etc.

If the keyboard or mouse are too far away, the user needs to stretch to reach them. This means the muscles in the shoulder and arm need to work harder to keep the user's hand in the correct position – causing strain, fatigue, and discomfort.

To help prevent harm, the user must make sure their equipment is set up correctly for them. The line of sight should be to the top of the monitor and the keyboard should be no more than 150mm from the user. The user should not have to stretch to reach their mouse – it should be a similar distance as the keyboard.

Part 3: Use health and safety procedures and solutions [PC2.3]

Assessor Checklist

Tick each item the learner has demonstrated or for which they have provided evidence of completion.

NOTE: Evidence may be gather over time from a range of different situations including, for example, the use of digital tools to complete another Unit Standard; the use of digital tools to complete work for an organisation; etc.

Item	✓
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IMPORTANT NOTE

Safe work practices are intended to address potential injury or harm from:

- furniture (eg desks, chairs)
- the environment (eg light, noise, temperature, air quality)
- the placement of digital device components (eg mouse, keyboard, screen).

1. The learner demonstrated the following **safe work practice**	✓

Hazard addressed: *Workstation*

Assessor Comments (eg give examples of how **safe work practices** were applied by learner, etc).

The learner ensured their workstation (desk, chair, etc) and digital device components were ergonomically arranged – allowing them to work in a neutral position. This included changing the height of the monitor to ensure comfortable eye gaze, and positioning the mouse and keyboard to ensure they were comfortably in reach.

The learner adjusted their chair (slightly higher) to make sure they could sit comfortably with their feet on the floor and knees bent roughly at right angles. The learner also adjusted the chair back to be more upright (so that they weren't leaning back – causing strain on their back).

The learner regularly changed their position while working at their desk.

See attached photos showing arrangement of furniture, digital device components, as well as the learner and environment (eg light sources).

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Item	✓
2. The learner demonstrated the following **safe work practice**	✓
Hazard addressed: *Lighting*	
The learner found that the light from a window was reflecting on their screen. They were able to turn their desk side on so the light source was coming from the side.	
3. The learner demonstrated the following **safe work practice**	✓
Hazard addressed: ***Noise***	
Assessor Comments (eg give examples of how **safe work practices** were applied by learner, etc) *The learner found that the noisy printer was causing some stress because it sometimes made it difficult to hear what people were saying on the phone/video conference calls, and was also an annoying ongoing noise. The learner discussed this with their supervisor and the supervisor arranged for the printer to be moved further away from the workstations.*	

END OF ASSESSMENT TASK